Ethereum & Smart Contract

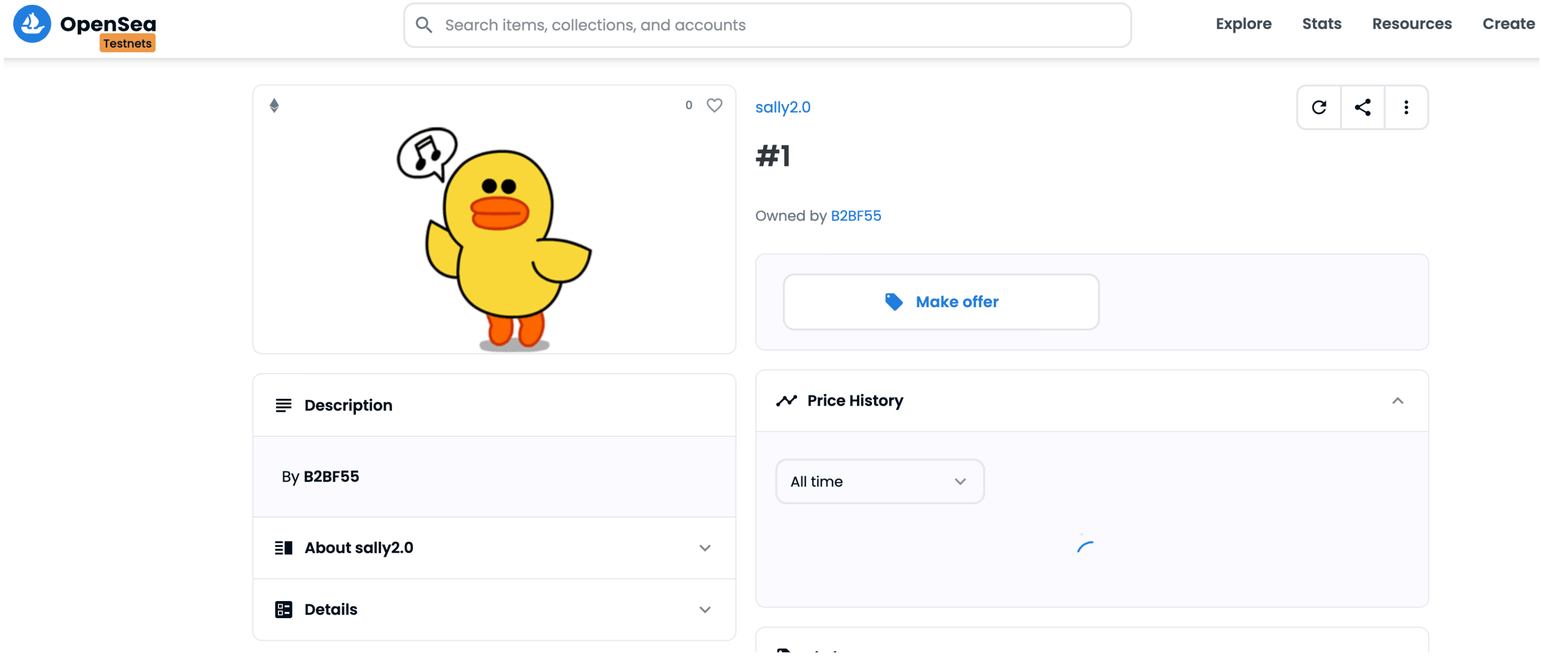
Alan Tseng

Outline

- Ethereum Basics
  - crypto wallet
  - EVM
  - Standards
- DApps
- Smart Contracts
  - Solidity
- Mint your NFT!







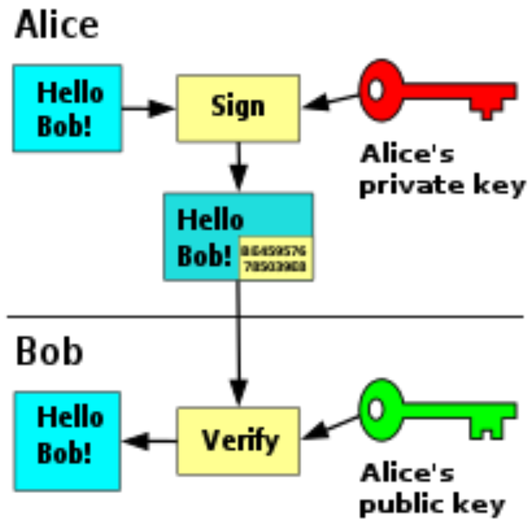
https://testnets.opensea.io/assets/rinkeby/0x6eabee10ec953096e05ef5718c4cb91df778416b/1

Before we start

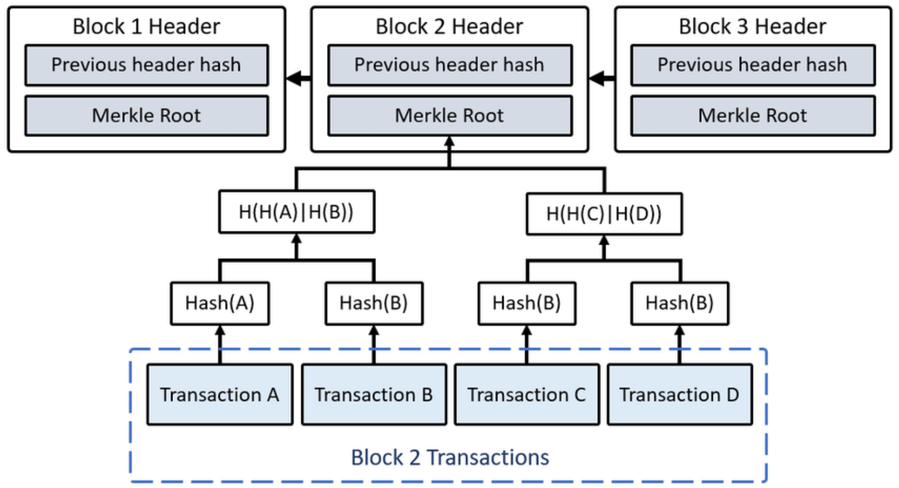
- Go to https://metamask.io/ & install
- Create an account and write down your mnemonic phrase
- Go to Advance, open “Show test networks” and select Rinkeby testnet
- Go to https://faucets.chain.link/rinkeby to get test ETH



Crypto wallets (recap)

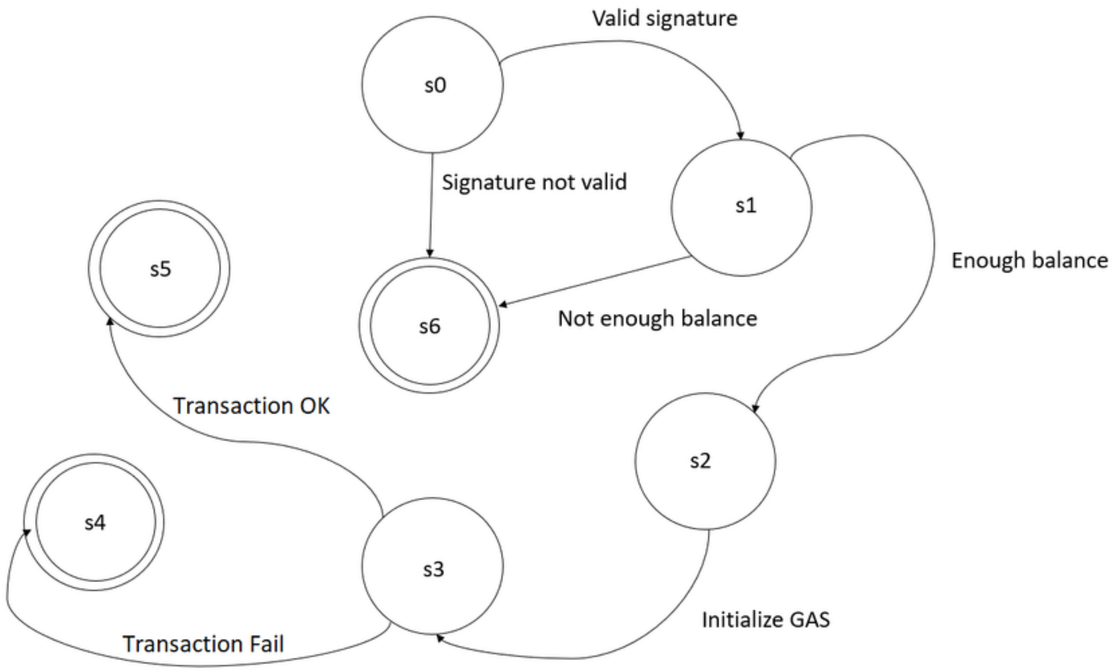


digital signature



Merkle tree in blockchain

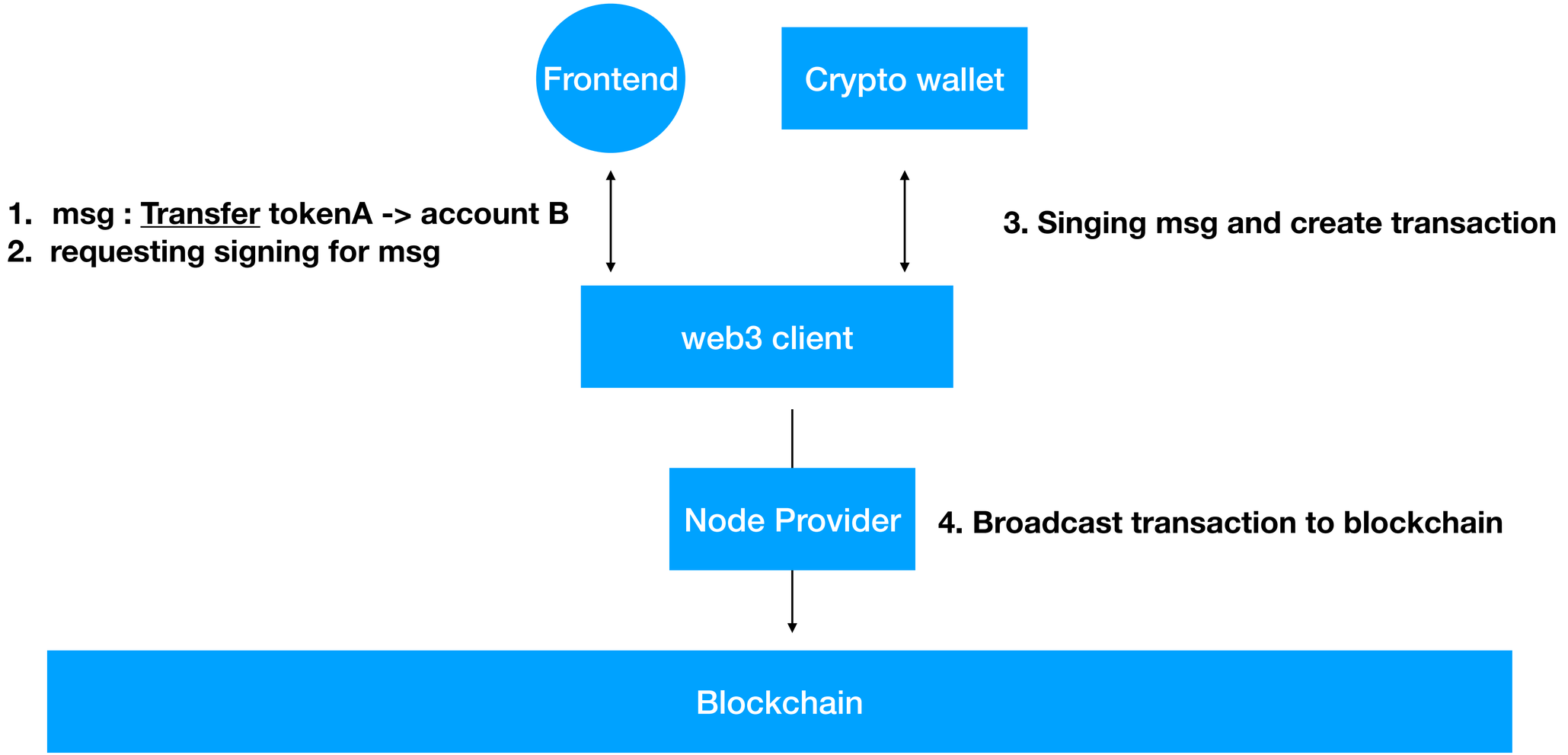
Crypto wallets (recap)



Ethereum state machine



Crypto wallets

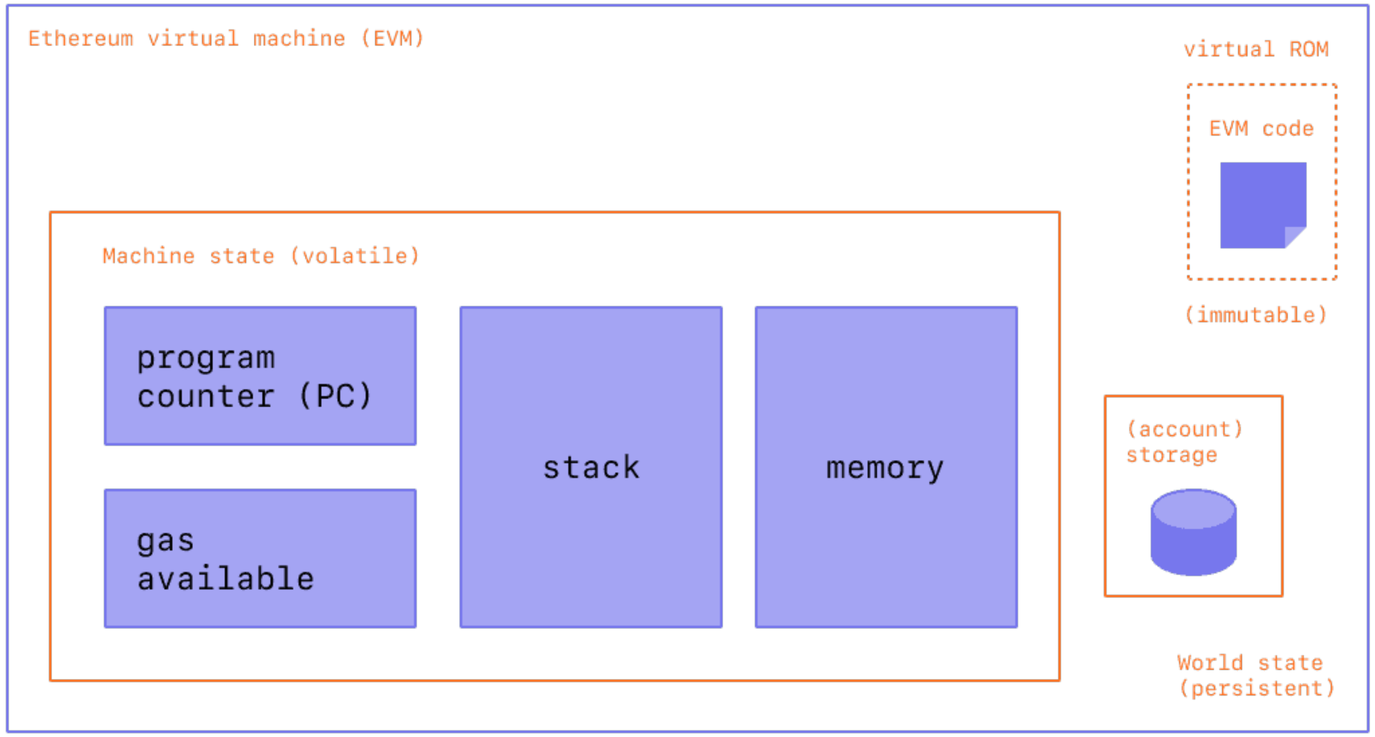


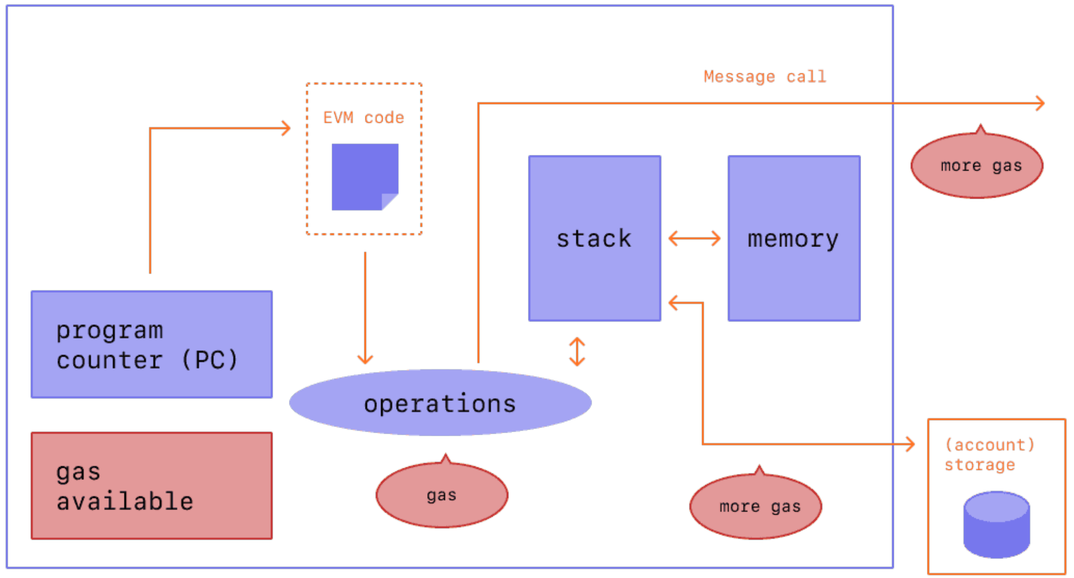
Glossary

- Wei: 10^-18 ETH
- GWei: 10^-9 ETH, Human-readable gas fees
- Gas: a unit of measuring the computational effort of conducting transactions
- Fee: A payment for using the blockchain to transact.



EVM


Ethereum virtual machine (EVM)
virtual ROM
EVM code
(immutable)
Machine state (volatile)
program counter (PC)
gas available
stack
memory
(account) storage
World state (persistent)


Message call
EVM code
more gas
stack
memory
program counter (PC)
operations
gas available
gas
more gas
(account) storage

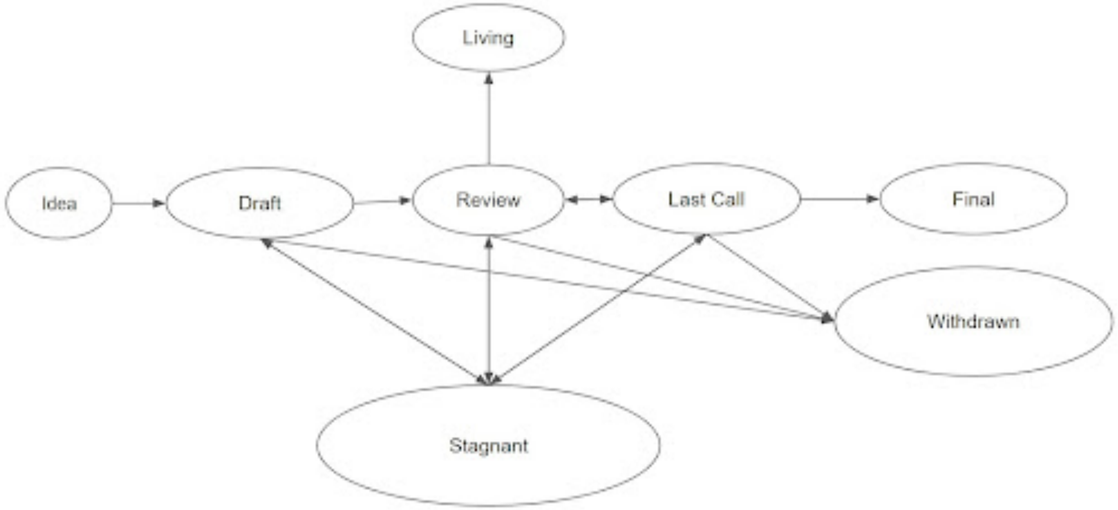
EVM - OPCODES

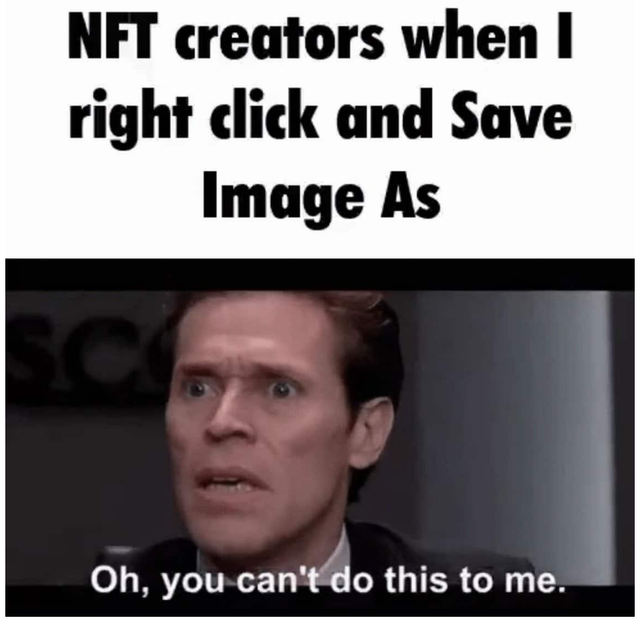
Stack	Name	Gas	Initial Stack
00	STOP	0	
01	ADD	3	`a, b`
02	MUL	5	`a, b`
03	SUB	3	`a, b`
04	DIV	5	`a, b`
05	SDIV	5	`a, b`
06	MOD	5	`a, b`
07	SMOD	5	`a, b`
08	ADDMOD	8	`a, b, N`
09	MULMOD	8	`a, b, N`
0A	EXP	A1↗	`a, b`
0B	SIGNEXTEND	5	`b, x`
0C-0F	*invalid*		



Ethereum - Standards

- EIP, Ethereum Improvement Proposals
  - ERC, Ethereum Request for Comment
    - ERC-20: Fongible Tokens
    - ERC-721: Non Fongible Tokens





DApps

Pros	Cons	Limitations
- Resistance to censorship - Verifiable behavior	- Network congestion - Maintenance	- Cost - UX

💡 **npx create-eth-app app**

Decentralized Applications

Finance

- Aave
- Uniswap
- dYdX

Arts Collectibles

- OpenSea
- CryptoPunks

Gaming

- Decentraland
- Axie Infinity

Technology

- Ethereum Name Service

Smart Contracts

- Limitation
  - Can not interact with real world
  - Maximum contract size: 24KB
- High-level language
  - Solidity: Gavin Wood (2014) -> Ethereum Foundation
  - Vyper
- Low-level language
  - Yul / Yul+



Smart Contracts - Tools

- https://remix.ethereum.org/: An online IDE for solidity
- https://etherscan.io/: for retrieve data on ethereum
- https://docs.openzeppelin.com/contracts/: A library for developing smart contract easier

Solidity syntax

Licence (optional)

Compiler version

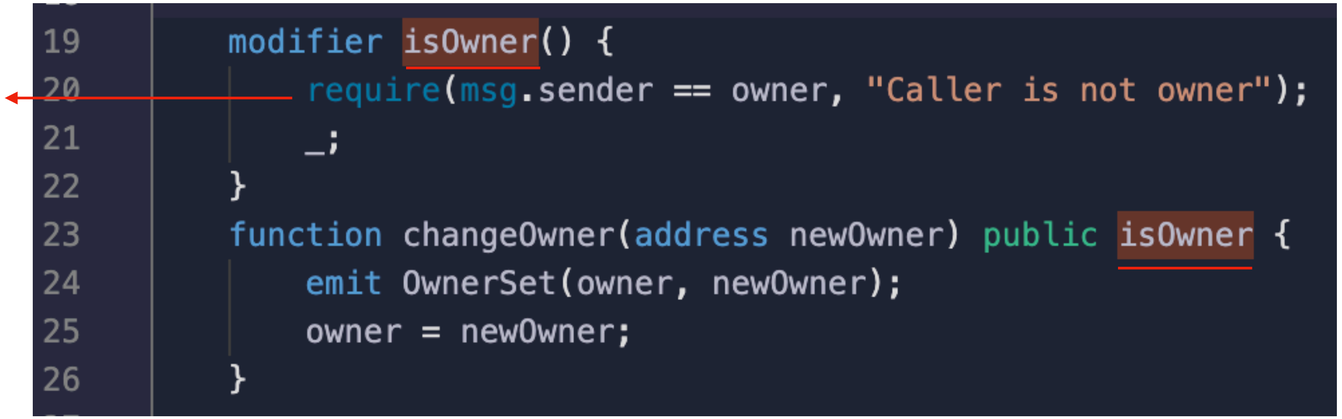
Contract

```
1  // SPDX-License-Identifier: GPL-3.0
2  pragma solidity >=0.7.0 <0.9.0;
3  contract Storage {
4      uint256 number;
5      constructor() {
6          number=1;
7      }
8      function store(uint256 num) public {
9          number = num;
10     }
11     function retrieve() public view returns (uint256){
12         return number;
13     }
14 }
```

Visibility

Solidity syntax - Function Modifier

If false,
revert all states and return remaining gas



```
19      modifier isOwner() {
20          require(msg.sender == owner, "Caller is not owner");
21          _;
22      }
23      function changeOwner(address newOwner) public isOwner {
24          emit OwnerSet(owner, newOwner);
25          owner = newOwner;
26      }
```

Make your own NFT!



- Remix IDE: https://remix.ethereum.org/
- Contract example: https://docs.openzeppelin.com/contracts
- Token uri: https://alanhc.github.io/nft/opensea-example.json
- Checkout NFT: https://testnets.opensea.io/{your_address}

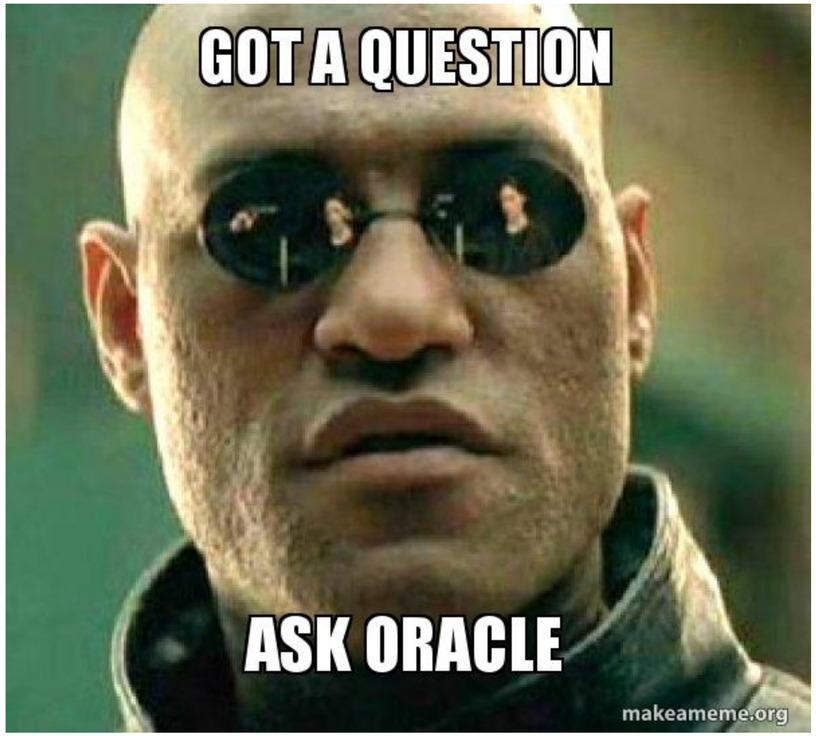
Summary

- What is Ethereum and how does it work
- What is a crypto wallet and how does it interact with blockchain
- What is DApps and its Opportunity
- What is smart contract and how to development

Sharing

- https://speedrunethereum.com/

Q&A



Q&A

Q1. 一個NFT項目會自己寫智能合約？

A. 分為兩類

1. 一般創作者，這些人會透過NFT交易平台，『鑄造』自己的NFT，以OpenSea鑄造NFT的功能屬於此類。

2. NFT項目方，通常為一個團隊，此類團隊通常會由智能合約工程師、創作者、行銷組成，遵循Token標準（ERC-20/ERC-721）使其可以在其他交易平台販售。

Q2. ERC 與 EIP的差異

A: Ethereum上的node上跑的程式都遵循EIP的準則，開發者會先提ERC草案，之後可能會被統整成EIP。

延伸閱讀：https://www.coindesk.com/learn/what-are-eip-and-erc-and-how-are-they-connected/



Thank you for your listening!